

PROJECT PROPOSAL
GROUP PROJECT
CST 291-2

Group No: CST 19

Pharmacy Inventory and Ordering System

Computer Science and Technology

Department of Computer Science and Informatics,
Faculty of Applied Sciences,
Uva Wellassa University of Sri Lanka

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DECLARATION

We hereby declare that the project will be developed by us with our own effort and that no part will be plagiarized without citations under the supervision of Mr. U.E. Ranasooriya. This Project Proposal is submitted for the partial fulfilment of the requirement of the course unit CST 219-2, Project 1, for the Computer Science and Technology degree program.

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1. Introduction

1.1 Project Title

Pharmacy Inventory and Ordering System

1.2 Project Description

This project focuses on the design and development of a web-based B2B pharmacy supply management system that connects retail pharmacies with pharmaceutical suppliers. The system enables pharmacies to digitally place orders, monitor stock levels, and receive timely deliveries. The system also facilitates the connection with the medical agents for special medicine needs.

On the supplier side, the system provides tools to manage product listings, update inventory quantities, and process incoming orders efficiently. The system also integrates automated features such as inventory tracking, expiry monitoring, and notification alerts, ensuring smooth and reliable pharmaceutical distribution.

This platform connects medical agents with pharmacies and customers by enabling them to obtain specialized and high demand medicines such as chemotherapy drugs, which are not accessible through traditional pharmaceutical distribution channels. It also aims to replace traditional manual ordering methods with a more accurate, transparent, and scalable digital solution.

1.3 Background and Motivation

In the Sri Lankan pharmaceutical supply chain, the process continues to depend on traditional and manual communication methods, including phone calls, handwritten documentation, spreadsheets, and messaging applications. Although these methods have been used for many years. They often create significant operational challenges for pharmacies and medicine suppliers as manual systems increase the risk of human error, reduce efficiency, and negatively impact inventory control and medicine distribution processes.

According to the World Health Organization, poor inventory management and weak communication systems can directly contribute to medicine shortages, treatment delays, and the wastage of pharmaceutical products [1].

The rapid advancement of web technologies platforms provides significant opportunities to modify pharmaceutical supply chain operations. Centralized web-based systems can improve communication, maintain accurate real-time records, and reduce paperwork and human errors.

The proposed system aims to develop a centralized web-based platform that connects pharmacies, suppliers, and medical agents within a single digital environment, enabling pharmacies to search for medicines, place orders online, monitor stock availability, and maintain transaction records.

Another major motivation for this project is the difficulty of obtaining rare or critical medicines that are not readily available through traditional pharmaceutical distribution. In Sri Lanka, certain medicines are available only through selected distributors or specialized medical agents. Therefore, the proposed system aims to establish a secure and centralized connection between customers and verified medical agents to improve access to essential medicines while ensuring reliability.

1.4 Problem in Brief

The pharmaceutical supply management system in Sri Lanka faces several operational challenges due to manual processes and traditional communication methods. Pharmacies usually order medicines through phone calls or messaging applications, which may lead to miscommunication, order errors, and delays in urgent medicine orders. In addition, many pharmacies depend on only a limited number of regular suppliers. The lack of real-time stock of information makes it difficult to compare prices and expiry dates. Rare drug orders take more time because these medicines are not easily available from a single supplier. In most cases, pharmacies need to contact several different suppliers to check availability. This process causes delays in medicine orders.

Another major problem is that when medicines are ordered through phone calls, pharmacies cannot verify expiry dates. Ordering medicines through phone calls often

lacks proper documentation, which can lead to overstocking. In advance, which often leads to the supply of near expiry or expired stock. Without proper monitoring and communication, pharmacies may stock too many medicines that expire before use, leading to waste, while at the same time, they may run out of important medicines when they are needed most. Furthermore, rare medicines are often not directly available in pharmacies, and expensive medicines must be purchased through medical agents. Many patients find it difficult to contact these agents and often do not have clear information about drug prices.

1.5 Proposed Solution

To address the existing challenges in pharmaceutical supply management in Sri Lanka, a web-based application is proposed to create a centralized digital platform connecting pharmacies, customers, suppliers, and medical agents. This system aims to replace manual and fragmented processes by digitizing the entire ordering and inventory process. The proposed system ensures better coordination between all stakeholders and reduces dependency on traditional communication methods such as phone calls and informal messaging.

One of the key features of the system is real-time medicine availability checking. This allows pharmacies to view updated stock information from suppliers before placing orders. This reduces communication errors, speeds up order processing, and improves transparency between suppliers and pharmacies. The system also includes real-time inventory management for pharmacies, enabling them to continuously monitor stock levels and expiry dates. In addition, the platform will provide order tracking and complete order history management. Another important feature is the integration with verified medical agents for handling special cases and rare medicine requirements. This ensures that even hard to find or urgent medicines can be sourced efficiently through a wider professional network.

Through those digitalized web application features can help reduce overstock and medicine waste while improving good communication between suppliers. This proactive approach significantly enhances inventory control and minimizes financial losses caused by expired or excess medicines.

1.6 Project Aim and Objective

1.6.1 Aim:

To develop an efficient and user-friendly web-based platform that serves as a channel between pharmacies, suppliers, and medical agents, facilitating streamlined pharmaceutical ordering and inventory management.

1.6.2 Objectives:

- 1) To design a user-friendly system with secure role-based access, including registration, login, password recovery, and admin controlled license verification for pharmacies, companies, medical agents, and customers.
- 2) To develop an efficient medicine inventory and order management system that enables companies to manage stock, track expiry dates, process orders, and allow pharmacies to place, track, and cancel orders.
- 3) To implement an integrated payment, billing, and notification system that supports multiple payment methods, invoice generation, payment tracking, and real time alerts for orders, inventory changes, and system activities.
- 4) To provide reporting, analytics, and administrative control features that allow system monitoring, sales visualization, commission management, user management, and policy enforcement for better decision-making

2. Methodology

2.1 Software Development Methodology

The project will use the Waterfall model as its software development method because this model provides a clear and organized development path. The Waterfall model works well for this project because it has established requirements for its web-based system, which includes product management, order processing and tracking, inventory management, and alert functions.

The project has already determined all system requirements, which will require only minor changes throughout the development process. The project team can complete all

project phases because this method enables better documentation, simpler management, and precise task distribution among team members.

The Waterfall model enables effective project planning and time management, which helps to complete the project within the given academic deadline requirements.

2.2 Application of the Waterfall Model in Pharmacy Supply Management System

Phase 1: Requirement Definition

The first phase of the project started with system requirements identification through user needs analysis, which included pharmacy owners, pharmaceutical suppliers, medical agents, and system administrators. The team used informal discussions together with their observations to identify the main problems that occur in the current manual system, including ordering delays, stock visibility issues, and expiry date tracking difficulties.

Activities carried out:

The team held informal discussions with pharmacy users, suppliers, and students who have knowledge about pharmacy inventory systems in order to identify the issues in current pharmacy inventory management processes.

Through those discussions, the team discovered several key problems, which included: the need to perform the ordering process manually, the inability to provide updates about stocks in real time, difficulty in managing expiry dates of medicines, the lack of capabilities for generating reports, and challenges in purchasing special medicines.

Based on the above-identified problems, the team determined the main system functions required for the proposed system. The system will provide an online ordering facility, support inventory management, notify users when stock levels reach low quantities, send notifications about product expiry dates, and generate an analysis of sales according to the past records.

The Waterfall model considers requirement engineering to be an essential stage, which enables developers to define all system features before they start building the software system.

Phase 2: System Design

The system architecture and design were created according to the collected requirements. The current stage of the project requires both database design work and system modelling to create a proper implementation plan.

Design components:

1. Database Design: A MySQL database schema created for
 - 1) Users
 - 2) Products
 - 3) Orders
 - 4) Inventory
 - 5) Payments
2. User Interface Design: Wireframes designed for
 - 1) Pharmacy Dashboard
 - 2) Supplier Dashboard
 - 3) Medical Agent Connection Portal
 - 4) Admin Panel
3. System Modelling: UML diagrams created,
 - 1) Use Case Diagram
 - 2) Class Diagram
 - 3) Sequence Diagram
4. Security Design:
 - 1) Role-based access control
 - 2) Secure login system

The Waterfall model requires system design completion before implementation begins because this method maintains a clear structure throughout all system components.

Phase 3: Implementation

The system development process follows design specifications during this phase. The development team uses module-based development to achieve better project organization while simplifying the debugging process.

Technologies used:

- 1) Frontend: React.js
- 2) Backend: PHP
- 3) Database: MySQL

Modules implemented:

- 1) User authentication module
- 2) Product management module
- 3) Order processing and tracking module
- 4) Inventory management module
- 5) Notification system

Each module is based on the specific user workflows. The system automatically updates inventory levels when a pharmacy order is placed while generating alerts for low stock situations.

The Waterfall model ensures that each requirement is translated into a functional component during this phase.

Phase 4: Testing

The system requires testing after implementation to confirm its functional accuracy and dependable performance.

Testing types:

- 1) Unit Testing: Testing each module, such as login, order processing, and inventory updates
- 2) Integration Testing: Ensuring that modules work together (ex, order placement updates inventory correctly)

- 3) System Testing: Verifying that the entire system meets the defined requirements

Key functionalities tested:

- 1) User registration and login
- 2) Product management
- 3) License and verification management
- 4) Order placement and tracking
- 5) Inventory updates
- 6) Low stock and expiry alerts

Testing ensures that the system functions operate as expected before the deployment.

Phase 5: Deployment

The system will enter its operational phase after testing achieves successful results.

Deployment activities:

- 1) Hosting the web application on a server
- 2) Configuring the domain and database
- 3) Migrating data from development to production
- 4) Final system validation process

The deployment phase ensures that the system is accessible to the users and operates smoothly in a real world environment.

Phase 6: Maintenance

The system requires constant observation with ongoing maintenance from the moment of its deployment.

Maintenance activities:

- 1) Fixing bugs and errors
- 2) Updating system features
- 3) Improving performance

4) Enhancing security

User feedback will drive system improvements, which will occur during upcoming software updates.

The Waterfall model defines a distinct maintenance phase that guarantees that systems remain usable while they maintain their operational reliability.

3. Requirements Identification

3.1 Functional Requirements

1. User Management

- 1) The system shall allow users to register as a pharmacy, company, medical agent, or customer by providing personal and business information.
- 2) The system shall enable users to log in with different roles, such as pharmacy, company, medical agent, customer, and admin.
- 3) User access should be authenticated by the system using a username and password.
- 4) The system shall support password recovery functionality for users who forget their login credentials.
- 5) The system shall provide membership plans with flexible billing options, including weekly, monthly, and yearly subscriptions, for pharmacies, companies, and medical agents.

2. License and Verification Management

- 1) The system should allow pharmacies, companies, and medical agents to upload their licenses or legal documents for verification to ensure authenticity before gaining full access to the platform.
- 2) The admin should have the authority to review, verify, and approve submitted licenses.

3. Inventory (Medicine) Management

- 1) The system shall allow users to view and browse available medicines through details such as name, price, and availability.
- 2) The system shall allow companies to add, update, and view medicine details, including brand name, generic name, price, expiry date, dosages, and batch number.
- 3) The system shall automatically update inventory stock levels when an order is placed.
- 4) The system shall send alerts to the companies when the product stock levels are low.

4. Order Management

- 1) The system shall allow pharmacies to add medicines to a cart and place orders online.
- 2) The system shall allow pharmacies to cancel orders.
- 3) The system shall allow companies to confirm orders.
- 4) The system shall store and maintain a complete history of all orders placed, including order details and status updates.
- 5) The system shall allow suppliers to view incoming orders and approve or reject them.
- 6) The system shall generate invoices for each completed order.
- 7) The system shall provide pharmacies with an order tracking feature to monitor the status of their orders.
- 8) The system shall allow companies to manage and update order status.
- 9) The system shall display the contact details of medical agents to customers when they request special medicine through an agent.

5. Payment and Billing System

- 1) The system shall generate an invoice and display a notification confirming the order approval by the company to support payment verification.

- 2) The system shall allow pharmacies to make payments through available methods such as bank transfer or online payment.
- 3) The system shall provide an option to upload a bank receipt for a payment made through bank transfer.
- 4) The system shall track the payment status, such as paid and pending.
- 5) The system shall generate a bill after payment is completed.
- 6) The system shall allow suppliers to view payment history for all transactions.

6. Notification System

- 1) The system shall display notifications on the dashboard of each user.
- 2) The system shall be able to generate notifications for key activities.
 - Order confirmation
 - Order progress changes
 - Inventory warnings.

7. Reporting and Analytics

- 1) The system shall allow admin users to monitor overall system performance.
- 2) The system shall present monthly sales data using graphs for better visualization.

8. Admin Panel

- 1) The system shall provide administrators with functionality to manage all registered users.
- 2) The system shall enable the admin to review and approve licenses submitted by companies, pharmacies, and medical agents before activating their accounts.
- 3) The system shall generate overall system reports.
- 4) The system shall display rules and regulations to companies, pharmacies, customers, and agents during the sign-in process to ensure awareness of platform policies before accessing the system.

- 5) The system shall provide the admin with functionality to manage commissions from companies.

3.2 Non-Functional Requirements

1. Usability

- 1) The system shall have a simple and user-friendly interface.
- 2) The system shall enable easy navigation for all types of users.
- 3) Pages shall load within an acceptable response time (e.g., under 3 seconds).

2. Performance

- 1) The system shall support multiple users simultaneously.
- 2) The system shall process data (orders, payments, updates) with minimal processing delay.

3. Scalability

- 1) The system shall support future expansion (more users, more products).
- 2) The system shall allow adding new features without affecting existing functionality.

4. Security

- 1) The system shall implement secure user authentication.
- 2) The system shall encrypt all user passwords.
- 3) The system shall prevent unauthorized access.
- 4) The system shall verify all licenses.

5. Reliability

- 1) The system shall operate continuously with high reliability and minimal system failures.
- 2) The system shall ensure accurate and consistent data storage.

6. Maintainability

- 1) The system code shall be well-structured to support easy maintenance and updates.
- 2) The system shall be designed to support efficient identification and fixing bugs and errors.

7. Availability

- 1) The system shall be available 24/7 with minimal scheduled downtime. Users shall be notified in advance of maintenance activities.

3.3 System Requirements

1. Hardware Requirements

- 1) Internet connection (Wi-Fi or mobile data).
- 2) Laptop or Desktop computer with:
 - Hard Disk: 10 GB or higher
 - RAM: 8 GB or higher
 - Processor: Intel i5 or higher
 - Operating System: Windows 10/11 or macOS
- 3) Storage device for the backups and project files.

2. Software Requirements

- 1) IDE: Visual Studio Code
- 2) Web Browser: Google Chrome / Microsoft Edge
- 3) Database Server: MySQL
- 4) Local Server: XAMPP / Apache / Node.js runtime
- 5) Design Tools: Figma

3. Technologies

- 1) Frontend: React.js
- 2) Backend: PHP
- 3) Database: MySQL
- 4) Version Control: Git and GitHub

3.4 User Roles

1. Admin

- 1) Manage user accounts, including creation, updating, and deletion of users.
- 2) Verify and approve pharmacy, supplier, and medical agent licenses.
- 3) Monitor system activities and user transactions.
- 4) Generate analytical reports related to users and financial activities.
- 5) Manage financial operations, including payments, commissions, and settlements.
- 6) Configure and maintain system settings, including roles and permissions.

2. Supplier

- 1) Register and authenticate within the system.
- 2) Manage company profile and business information.
- 3) Add, update, and remove pharmaceutical product listings.
- 4) Manage inventory levels and stock availability.
- 5) Receive, review, and respond to incoming orders from pharmacies.
- 6) Update the order processing stage and the delivery stage.
- 7) Review and respond to ratings and feedback submitted by pharmacies.

3. Pharmacy

- 1) Register and authenticate within the system.
- 2) Maintain and update pharmacy profile.
- 3) Browse, search, and filter available pharmaceutical products.
- 4) Place and manage orders through the system.
- 5) Track the status of orders in real time.
- 6) Receive notification regarding stock availability, order updates, and system alerts.
- 7) Provide ratings and reviews for suppliers and products.

3.4.4 Customer

- 1) Register and authenticate within the system.
- 2) Browse and search available medicines.
- 3) Request medicines from agents.
- 4) Submit feedback, ratings, and reviews for medical agents.

3.4.5 Medical Agent

- 1) Register and authenticate within the system.
- 2) Maintain and update personal profile information.
- 3) Add, update, and manage specialized or rare pharmaceutical products.

4. Project Plan

4.1 Project Schedule Overview

The project will be completed over a period of 14 weeks, following the Software Development Life Cycle (SDLC). Each phase is carefully planned to ensure timely completion.

4.2 Gantt chart

Tasks/Activities	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Requirement Gathering														
Requirement Analysis and Specification														
Project Proposal Submission														
Project Proposal Presentation														
ER Diagram and Use Case Diagram														
Database Design														
Backend Development														
Frontend Development														
Implementation and Unit Testing														
Progress Presentation														
Integration and System Testing														
Operation and Maintenance														
Documentation														

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